MineMind

JV Inkai: acid delivery timing and wellfield commissioning

Which new block groups start later when acid arrives late?

A source-informed, representative digital twin on OpenEgiz.
18 twins. One adjustable scenario. Complete reset.

Synthetic block groups and schedules. Assumed commissioning response.
No site calibration, optimizer or actual operating forecast.

Connected platform and mine flows

Model publisher -> MQTT -> Eclipse Ditto -> spatial interface

The UI reads all operating states and projections back from Ditto.
Missing twins or mixed revisions produce an error, not substitute model data.

Commissioning acid

Separate area stocks -> whole-dose reservation -> acidification -> commissioning

Uranium-bearing solution and product

Block extraction -> area IX -> recovered-U queue -> shared MPP drying

Existing/sustaining acid supply is protected outside this model. Satellite transfer is aggregated.

Evidence and explicit assumptions

Public facts

MPP / Sat-1 / Sat-2 IX: 2.7 / 6.3 / 4.5 Mlb U3O8/year.
Drying: 8.3 Mlb/year. Plant recovery: approximately 98%.

Synthetic exercise and assumptions

Nine group identities, positions, stocks, rates and schedules.
Whole commissioning doses: 100-160 t per new group.
Seven-day acidification, seven-day ramp, fixed 100 mg U/L.
Annual capacities converted to uniform daily model limits.

The operator reported new-block acidification delays despite 106% nine-month acid fulfilment. This supports the timing mechanism, not these coefficients or a 28-day reconstruction.

Sources: Cameco 2024 Inkai report, printed pp.80-82; Kazatomprom 3Q24 update; PARAMETERS.md

Accountable inventories and constraints

Acid stock = initial stock + deliveries - reserved doses

Extracted U = dried product U + queue U + unrecovered U

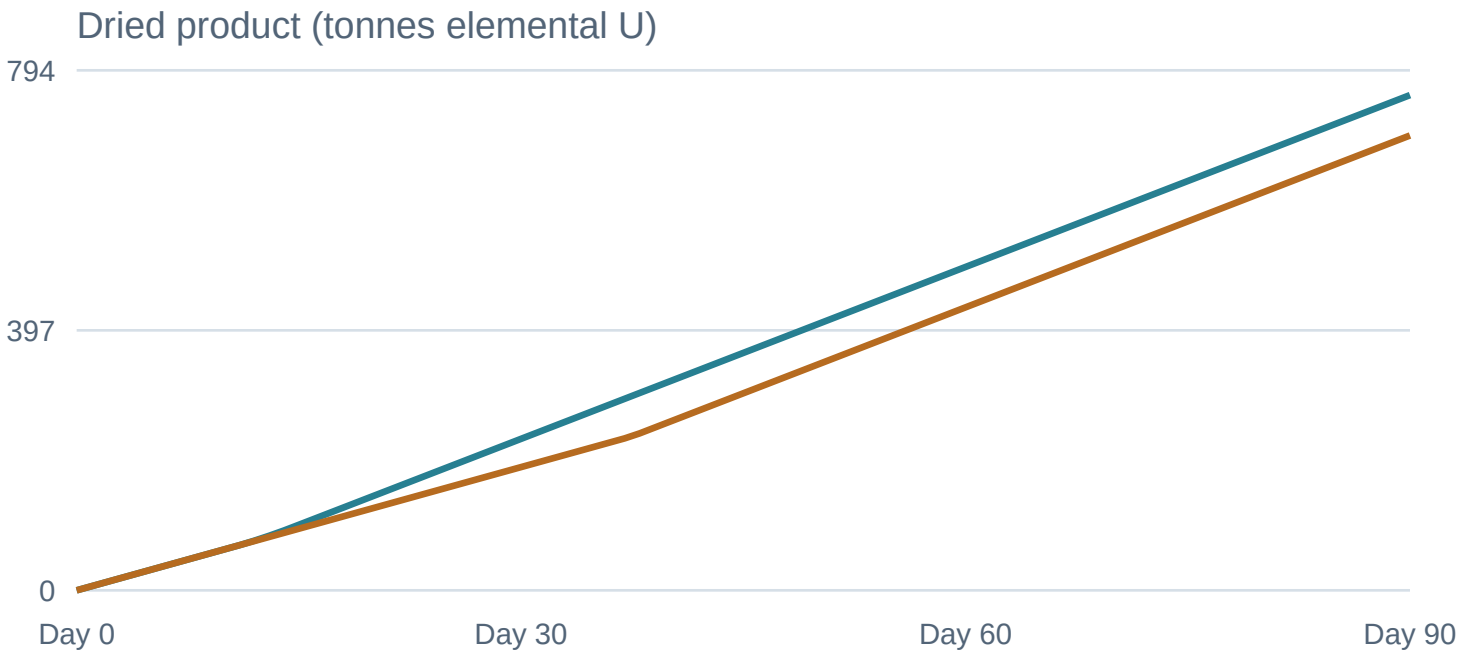
Initial group U = remaining U + cumulative extraction

Area IX limits proportionally throttle extraction.
Recovered uranium remains in the drying queue when packaging binds.
The 2% unrecovered stream is recorded; uranium is not destroyed.

Conservation tests establish internal consistency, not real-world operational validation.

Simulated delivery-delay comparison

First arrivals: day 3 -> day 31 (+28 days). Second arrivals stay day 60.



Baseline

28-day delay

Day 30: 231.30 vs 188.16 t U.
Day 90 difference:
-61.66 t U.

New B1 starts: D12 -> D38.
Same total acid by day 90.

A model outcome, not an Inkai loss estimate. Different assumptions or bottlenecks can change the difference.

Verification and reproduction

21 unit tests (Windows + Ubuntu)	PASSED
Live MQTT / Ditto / scenario / reset	PASSED
Original platform end-to-end smoke	PASSED
Browser spatial controls	PENDING
Fresh credentials and data-volume startup	PENDING

Start: sudo apt-get update && sudo apt-get install -y make
make inkai-test | make inkai | make inkai-smoke

Open localhost:8088. Reset -> view day 30 -> apply delay 28.
Inspect day 38, then day 90. Reset baseline to undo completely.

Local test: Ubuntu 24.04 WSL2 / Docker 29.1.3 / Compose 2.40.3 / about 7 GiB RAM.
Fresh-volume testing reuses downloaded images. Exact 16-GiB KVM judge reproduction remains unverified.
Detailed evidence, sources, assumptions and known limitations are committed alongside the code.